
The Collapse Model: Spectral Geometry and Auditable Data-Pool Screening

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Abstract

Given M frozen-representation data pools, which candidates can be short-listed without labels before expensive evaluation? We separate exact spectral claims from empirical screening. For fixed, label-independent preprocessing and declared source weights, the merged spectrum is exactly determined by a sum of positive-semidefinite Gram operators. Under explicit block-compatible singular directions this sum has an exact local 2×2 decomposition; a constructive counterexample shows that marginal spectra and principal angles—and hence the earlier four-scalar summary—do not determine merged effective rank in general. Our primary Stage-1 method is a composable rank- L PSD Gram sketch. Transmitted tail statistics give deterministic Fannes–Audenaert intervals for log-effective-rank error and a sufficient certificate for one Full-Gram greedy step. Controlled sweeps across four encoders show that all 2,880 selected-prefix intervals cover the exact score, but none of 7,200 greedy steps is certified for $L \leq 50$, and DPP is at least as competitive. In 28 held-out encoder-target units, even validation-oracle source-supervised adapters trail frozen identity by 0.067 (target-cluster 95% CI $[-0.098, -0.044]$); small shortlist regret is therefore not downstream benefit. We make no global-optimality, privacy, practical-compression, or downstream-benefit claim. Code is available at <https://anonymous.4open.science/r/collapsemodel-1DB6>.

1 Introduction

Modern pretrained models increasingly rely on combining features from many candidate datasets—multi-domain transfer learning, data curation for self-supervised pretraining [Gadre et al., 2023], and continual learning all face the question: given M candidate datasets, which $k \ll M$ subset preserves the diversity of the full pool when their features are concatenated? We call this the label-free data-pool compression problem: unlike classical coreset selection [Mirzasoileiman et al., 2020] and recent sample-level data curation methods (DSIR [Xie et al., 2023], DsDm [Engstrom et al., 2024], D² Pruning [Maharana et al., 2023]), we seek compression at the dataset level (not the sample level), without access to labels or downstream tasks. The criterion is representational diversity, formalized as the effective rank $r_{\text{eff}}(H) = \exp(-\sum_i p_i \log p_i)$ of the merged features [Roy and Vetterli, 2007, Garrido et al., 2023]. No prior work characterizes how dataset-level spectra and relative subspace geometry jointly affect r_{eff} under feature concatenation, forcing practitioners to rely on exhaustive search or heuristics that scale poorly with M .

Prior work addresses related but distinct problems. Spectral diversity metrics such as RankMe [Garrido et al., 2023], VNE [Kim et al., 2023], the Vendi Score [Friedman and

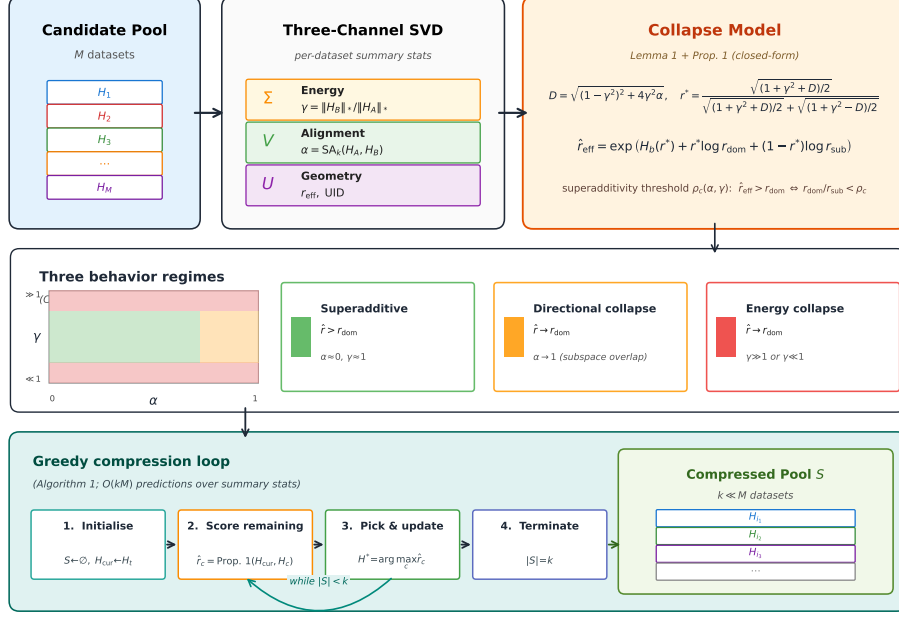


Figure 1: Spectral mechanism and selection abstraction. Top/middle: the paired-direction special case and Collapse-4S diagnostic; these panels explain local behaviour but are not a general guarantee. Bottom: the common greedy loop. Our primary selector scores this loop by composing rank- L PSD factors; Collapse-4S is retained as an ablation.

Dieng, 2023], and its generalized form [Pasarkar and Dieng, 2024] quantify the diversity of a single set via entropy of a singular-value or similarity-matrix spectrum; they do not predict how two sets’ diversity combines under concatenation. Label-free coreset selection (ELFS [Shen et al., 2025], ZCore [Griffin et al., 2024], DEFT-UCS [Das and Khetan, 2024]) selects a subset of samples using geometry or proxy dynamics; it is complementary but orthogonal to our dataset-level formulation. Pool-level curation (DataComp [Gadre et al., 2023], DataComp-LM [Li et al., 2024], DsDm [Engstrom et al., 2024], Quad [Zhang et al., 2025]) uses filtering or model-aware heuristics without a closed-form diversity criterion. Representation similarity measures (SVCCA [Raghu et al., 2017], PWCCA [Morcos et al., 2018], CKA [Kornblith et al., 2019], PARC [Bolya et al., 2021]) compare two representations but do not predict merged rank. Transferability estimators (LogME [Zhang et al., 2021], H-score [Bao et al., 2019], TransRate [Huang et al., 2022], LEEP [Li et al., 2021], reassessed by Ibrahim et al. [2022]) and PMI-based dataset valuation [Zhang et al., 2024] predict downstream accuracy but either require labels or do not model merging. Task2Vec [Achille et al., 2019] avoids labels but requires probe fine-tuning. Model-merging methods (Task Arithmetic [Ilharco et al., 2023], TIES-Merging [Yadav et al., 2023], DARE [Yu et al., 2024]) combine parameters, not features. Intrinsic-dimensionality work [Pope et al., 2021, Ansuini et al., 2019] characterizes individual matrices but not merged ones. To our knowledge, no existing work combines an exact local spectral analysis of $r_{\text{eff}}([H_A; H_B])$ with a summary-statistic predictor for dataset-level selection.

We start from the exact identity $G_{A \cup B} = G_A + G_B$ and keep two approximations conceptually separate. The earlier four-summary predictor, which we call Collapse-4S, compresses a paired-direction model into $(r_A, r_B, \gamma, \alpha)$. It is useful for mechanism diagnostics and as an empirical baseline, but Proposition 5 proves that these summaries are not sufficient in general. The primary computational method, Collapse-Gram- L , instead transmits a rank- L PSD factor and certified tail statistics for every source. These summaries compose by addition, retain spectral pairing within the retained subspace, and provide a deterministic error interval (Theorem 1).

Why summary-based screening? Exhaustive search requires $\binom{M}{k}$ merged decompositions and access to all feature rows. Collapse-Gram- L uses $O(kM)$ candidate evaluations and communicates $O(Ld)$ values plus three tail statistics per source. It does not require raw sample features at the coordinator, but this compression is not a differential-privacy guarantee.

Contributions. (1) Exact algebra and an information limit: an unconditional Gram representation, an exact block-compatible decomposition, and an explicit counterexample to low-dimensional identifiability (Propositions 1–5). (2) Certified compression: a composable rank- L PSD sketch with a tail-dependent log-effective-rank interval and a sufficient certificate for matching one step of Full-Gram greedy (Theorem 1, Corollary 6). (3) An audited two-stage protocol: label-blind Stage-1 sampling, declared equal-source weighting, target-isolated Stage-2 evaluation, matched overlap diagnostics, source-independent controls, and provenance-checked resume semantics (Sec. 5).

Empirical scope. The existing pairwise experiments evaluate Collapse-4S, not the new rank- L guarantee. They show strong ordering correlation on the evaluated pairs, but only a 0.003 Spearman improvement over $r_A + r_B$ and encoder-dependent calibration. Historical pool experiments likewise show that spectral methods often select similar clean-pool subsets. We retain these results as bounded evidence and report corrected rank- L , matched overlap, and task-aware comparisons separately rather than treating the old tables as validation of Theorem 1.

2 Spectral Background

For a feature matrix $H \in \mathbb{R}^{N \times d}$ with compact SVD $H = U\Sigma V^\top$, let $\Sigma = \text{diag}(\sigma_1 \geq \dots \geq \sigma_r > 0)$.

Σ channel — spectral energy. The effective rank summarizes spectral distribution:

$$r_{\text{eff}}(H) = \exp\left(-\sum_{i=1}^r p_i \log p_i\right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}, \quad (1)$$

where p_i is the singular-value share (not squared), so $r_{\text{eff}} \in [1, r]$. The nuclear norm $\|H\|_* = \sum_i \sigma_i$ is the normalizing mass. In all numerical results, we retain singular modes satisfying $\sigma_i > \sigma_1 \sqrt{8\epsilon_{64} \max(N, d)}$. Direct-SVD and Gram-eigensolver paths apply this same cutoff before computing nuclear mass or effective rank.

V channel — directional structure. Right singular vectors V encode which directions are activated.

Definition 1 (Subspace Alignment). Let $\hat{V}_A^k, \hat{V}_B^k \in \mathbb{R}^{d \times k}$ be the top- k right singular vectors of H_A, H_B . The Subspace Alignment at level k is:

$$\text{SA}_k(H_A, H_B) = \frac{1}{k} \left\| \hat{V}_A^{k\top} \hat{V}_B^k \right\|_F^2 = \frac{1}{k} \sum_{i=1}^k \cos^2 \theta_i, \quad (2)$$

where $\theta_1 \leq \dots \leq \theta_k$ are the principal angles.

$\text{SA}_k \in [0, 1]$ summarizes top-subspace overlap, but discards the principal-vector basis and its coupling to singular values. This loss of information is central to Proposition 5.

3 The Collapse Model

3.1 Setup and Notation

Let $H_A \in \mathbb{R}^{N_A \times d}$ and $H_B \in \mathbb{R}^{N_B \times d}$ be L2-row-normalized feature matrices with effective ranks $r_A = r_{\text{eff}}(H_A)$, $r_B = r_{\text{eff}}(H_B)$, nuclear norms $\|H_A\|_*$, $\|H_B\|_*$, and subspace alignment $\alpha = \text{SA}_k(H_A, H_B)$. Define:

$$\gamma = \frac{\|H_B\|_*}{\|H_A\|_*} \in (0, \infty), \quad r_{\text{dom}} = \begin{cases} r_A & \gamma \leq 1 \\ r_B & \gamma > 1 \end{cases}, \quad r_{\text{sub}} = \begin{cases} r_B & \gamma \leq 1 \\ r_A & \gamma > 1 \end{cases}. \quad (3)$$

104 γ is the energy ratio of the two matrices; r_{dom} is the effective rank of the higher-energy
105 matrix.

106 All source preprocessing is fixed before subset selection. For source j with N_j rows, we
107 declare a total source weight $\beta_j > 0$ and use $X_j = \sqrt{\beta_j/N_j} H_j$. Our unequal-size default is
108 $\beta_j = 1$ (equal total mass per source); $\beta_j = N_j$ is the declared equal-per-sample alternative.
109 Although effective rank is invariant to a common scale, it is not invariant to changing relative
110 source weights.

111 3.2 Main Results

112 Our analysis separates three levels that must not be conflated: an unconditional Gram
113 identity, an exact paired-direction model under explicit decoupling assumptions, and a low-
114 dimensional summary predictor for general spectra.

115 Proposition 1 (Exact Gram Representation). Let $H_A = U_A \Sigma_A V_A^\top$ and $H_B = U_B \Sigma_B V_B^\top$ be
116 compact SVDs, and complete V_A to an orthogonal basis $Q_A = [V_A \ V_{A,\perp}]$. With $C = Q_A^\top V_B$
117 and $\tilde{\Sigma}_A^2$ denoting Σ_A^2 padded by zeros to dimension d , the merged Gram operator satisfies

$$Q_A^\top (H_A^\top H_A + H_B^\top H_B) Q_A = \tilde{\Sigma}_A^2 + C \Sigma_B^2 C^\top. \quad (4)$$

118 Consequently, the merged singular values are the square roots of the eigenvalues of the
119 right-hand side.

120 Equation (4) is unconditional. In general C is dense: a singular direction of one matrix can
121 couple to several directions of the other. Individual spectra and an average principal-angle
122 statistic therefore do not determine the merged spectrum.

123 Lemma 2 (Exact: 2×2 Gram Block Eigenstructure). Assume paired, decoupled directions:
124 for each i , $V_{B,i} = \cos \theta_i V_{A,i} + \sin \theta_i V_{\perp,i}$, where the two-dimensional planes $\text{span}\{V_{A,i}, V_{\perp,i}\}$
125 are mutually orthogonal. Define the mode-wise energy ratio $\gamma_i = \sigma_{B,i}/\sigma_{A,i}$ and $\alpha_i = \cos^2 \theta_i$.
126 The i -th Gram block has eigenvalues

$$\lambda_{\pm,i} = \frac{\sigma_{A,i}^2}{2} (A_i \pm D_i), \quad A_i = 1 + \gamma_i^2, \quad D_i = \sqrt{(1 - \gamma_i^2)^2 + 4\gamma_i^2 \alpha_i}. \quad (5)$$

127 The corresponding singular values are $\sqrt{\lambda_{+,i}}$ and $\sqrt{\lambda_{-,i}}$. This result is exact under the
128 stated pairing and decoupling conditions (Appendix B).

129 The global ratio $\gamma = \|H_B\|_*/\|H_A\|_*$ does not imply $\gamma_i = \gamma$. Replacing the collection $\{\gamma_i, \alpha_i\}$
130 by global summaries (γ, α) is an approximation step.

131 Proposition 3 (Exact Proportional-Spectrum Model). Under Lemma 2, further assume
132 $\sigma_{B,i} = \gamma \sigma_{A,i}$ and $\alpha_i = \alpha$ for every i . Let $A = 1 + \gamma^2$, $D = \sqrt{(1 - \gamma^2)^2 + 4\gamma^2 \alpha}$, and

$$q = \frac{\sqrt{(A+D)/2}}{\sqrt{(A+D)/2} + \sqrt{(A-D)/2}}. \quad (6)$$

133 Then the merged spectrum consists of two scaled copies of the spectrum of H_A , and

$$r_{\text{merged}} = r_A \exp(H_b(q)), \quad r_A = r_B. \quad (7)$$

134 Definition 2 (Collapse Summary Predictor). For general spectra, we compress the mode-wise
135 quantities to the global nuclear-norm ratio γ and mean top- k alignment α , compute q from
136 Eq. (6), and define

$$\hat{r}_{\text{merged}} = \exp\left(H_b(q) + q \log r_{\text{dom}} + (1 - q) \log r_{\text{sub}}\right), \quad (8)$$

137 where $H_b(p) = -p \log p - (1 - p) \log(1 - p)$.

138 Equation (8) is a theory-motivated low-dimensional approximation, not an identity or a
139 universally bounded estimator. It interpolates the input spectral entropies and adds the
140 binary entropy of the idealized two-branch split. Its accuracy on general spectra is an
141 empirical question evaluated in Sec. 4.

142 Interpretation. Equation (8) is a weighted log-geometric mean of the individual ranks plus
 143 a balance bonus $H_b(q)$ (maximal when $q=0.5$, vanishing as $q \rightarrow 0, 1$).

144 Behavior regimes. Complementary directions ($\alpha \approx 0, \gamma \approx 1$): $q \approx 0.5$ and $\hat{r}_{\text{merged}} \approx 2\sqrt{r_A r_B}$.
 145 At exactly $(\alpha, \gamma) = (0, 1)$, this prediction is superadditive if and only if $r_{\text{dom}}/r_{\text{sub}} < 4$.
 146 Directional collapse ($\alpha \rightarrow 1$): $q \rightarrow 1, H_b(q) \rightarrow 0, \hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$. Energy collapse ($\gamma \gg 1$ or $\gamma \ll 1$):
 147 $q \rightarrow 1, \hat{r}_{\text{merged}} \rightarrow r_{\text{dom}}$. When $\alpha \rightarrow 1$ and $\gamma \gg 1$, the merged rank can fall below r_{sub} —a net loss
 148 in diversity.

149 3.3 Superadditivity Threshold

150 Corollary 4 (Summary-Model Superadditivity Diagnostic). The summary predictor in
 151 Eq. (8) predicts $\hat{r}_{\text{merged}} > r_{\text{dom}}$ for $q < 1$ if and only if

$$\frac{r_{\text{dom}}}{r_{\text{sub}}} < \rho_c(\alpha, \gamma) \equiv \exp\left(\frac{H_b(q)}{1-q}\right), \quad (9)$$

152 where q is from Eq. (6). The threshold ρ_c is increasing in α : $\rho_c \geq 4.0$ for all $\alpha \in [0, 0.9]$ at
 153 $\gamma = 1$. As $q \rightarrow 1^-$, $\rho_c \rightarrow \infty$ while the predicted gain tends to zero. At the degenerate point
 154 $q = 1$ the division in Eq. (9) is undefined and Eq. (8) gives $\hat{r}_{\text{merged}} = r_{\text{dom}}$ exactly.

155 This is an exact decision boundary for the summary model, and for the proportional-
 156 spectrum model of Proposition 3; on general data it is a diagnostic rather than a guarantee.
 157 For finite-precision empirical decisions, a predicted or measured value x is declared superad-
 158 ditive only when $x - r_{\text{dom}} > 10^{-10} \max(x, r_{\text{dom}}, 1)$; this suppresses roundoff-only positives
 159 at the equality boundary.

160 Physical meaning. ρ_c is a decision threshold internal to the summary model. Near $q = 1$,
 161 it can classify an arbitrarily small positive entropy increment as superadditive; the mag-
 162 nitude $\hat{r}_{\text{merged}} - r_{\text{dom}}$ must therefore be reported alongside the binary diagnosis. Exact
 163 directional overlap gives $q = 1$ and no predicted gain. The proofs and model scope are in
 164 Appendices B, L.

165 3.4 Marginal Spectra and Principal Angles Are Not Sufficient

166 Proposition 5 (Non-identifiability from Marginal Geometry). Even the two marginal singu-
 167 lar spectra together with the complete set of principal angles between the right singular
 168 subspaces do not uniquely determine $r_{\text{eff}}([H_A; H_B])$. Consequently, neither does the tuple
 169 $(r_A, r_B, \gamma, \alpha)$.

170 Counterexample. Let e_1, e_2, e_3 be standard basis vectors and define the row-normalized
 171 matrices

$$H_A = [e_1, e_1, e_1, e_1, e_2]^\top, \quad H_B^{(1)} = [e_1, e_1, e_1, e_1, e_3]^\top, \quad H_B^{(2)} = [e_1, e_3, e_3, e_3, e_3]^\top. \quad (10)$$

172 All three have nonzero singular values (2, 1). Both right-subspace pairs have principal angles
 173 $\{0, \pi/2\}$, yet their merged Gram spectra are (8, 1, 1) and (5, 4, 1), respectively. The corre-
 174 sponding effective ranks are 2.626012 and 2.849536. Thus even complete marginal spectra
 175 and principal angles omit how spectral mass is assigned to coupled directions; the four scalar
 176 summaries discard still more information. $\square$

177 3.5 Composable Rank- L Gram Sketch

178 The counterexample motivates approximating the PSD operator itself rather than compress-
 179 ing it to averaged angles. For the compact SVD of each preweighted source, define

$$B_j = \Sigma_{j,1:L} V_{j,1:L}^\top, \quad \tilde{G}_j = B_j^\top B_j, \quad E_j = G_j - \tilde{G}_j \succeq 0. \quad (11)$$

180 Attach the tail statistics $\tau_j = \text{tr} \sqrt{E_j}$, $\omega_j = \text{tr}(E_j)$, and $u_j = \text{rank}(E_j)$; certified upper
 181 bounds may replace their exact values. For a subset S , vertically stack the factors into B_S .
 182 Then $\tilde{G}_S = B_S^\top B_S$, so its nonzero spectrum is obtained from the small core $B_S B_S^\top$ without
 183 reconstructing sample features.

184 For a PSD matrix G , write $\mathcal{R}(G) = \exp \mathcal{H}(\sqrt{\lambda(G)}/\text{tr} \sqrt{G})$; hence $\mathcal{R}(H^\top H) = r_{\text{eff}}(H)$.
 185 Define

$$T_S = \min \left\{ \sum_{j \in S} \tau_j, \sqrt{\min(d, \sum_{j \in S} u_j) \sum_{j \in S} \omega_j} \right\}, \quad \varepsilon_S = \frac{T_S}{\text{tr} \sqrt{\tilde{G}_S + T_S}}. \quad (12)$$

186 Theorem 1 (Rank- L Log-Effective-Rank Bound). Assume $\tilde{G}_S \neq 0$, and let $D_S =$
 187 $\min(d, \sum_{j \in S} \text{rank}(G_j))$. For $D \geq 2$, set

$$\Phi_D(\varepsilon) = \begin{cases} h_b(\varepsilon) + \varepsilon \log(D-1), & 0 \leq \varepsilon \leq 1 - 1/D, \\ \log D, & 1 - 1/D < \varepsilon \leq 1. \end{cases} \quad (13)$$

188 Then, without a block-compatibility assumption,

$$\left| \log \mathcal{R}(G_S) - \log \mathcal{R}(\tilde{G}_S) \right| \leq \Phi_{D_S}(\varepsilon_S). \quad (14)$$

189 For $D_S = 1$ the error is zero.

190 Corollary 6 (One-Step Full-Gram Certificate). At a greedy step, let $\hat{F}_j = \log \mathcal{R}(\tilde{G}_{S \cup \{j\}})$ and
 191 $e_j = \Phi_{D_{S \cup \{j\}}}(\varepsilon_{S \cup \{j\}})$. If a candidate j^* satisfies

$$\hat{F}_{j^*} - e_{j^*} > \max_{j \neq j^*} (\hat{F}_j + e_j), \quad (15)$$

192 then j^* uniquely maximizes the exact Full-Gram score in that round.

193 The interval accumulates tails from the selected set and the candidate; using only the
 194 candidate tail is invalid. Failure to separate intervals is a diagnostic to increase L or fall
 195 back to Full-Gram scoring. The certificate is local to one greedy step: effective rank is
 196 generally nonmonotone and nonsubmodular, so it gives no global subset-optimality claim
 197 (Appendix C).

198 Legacy Collapse-4S calibration. On the evaluated real-data pairs Eq. (8) systematically
 199 over-predicts. A fitted multiplicative constant reduces average scale error; the reported
 200 relation $c=0.946-0.175 \log(N/r_{\text{eff}})$ is an empirical fit over the evaluated encoder families,
 201 not a universal law (Appendix L).

202 4 Evaluation of the Legacy Four-Summary Predictor

203 This section reports the historical evaluation of Collapse-4S. These experiments test Eq. (8),
 204 not the rank- L approximation or its certificate. We organize the evidence as: (i) idealized-
 205 model agreement—the predictor agrees with data generated under its summary assump-
 206 tions (§4.1, synthetic grid); (ii) real-data association—the formula remains correlated with
 207 approximation across 5 encoder families, 14 vision datasets, and 19 text corpora (§4.2,
 208 427 pairs); (iii) behavioral regime agreement—empirical superadditivity patterns match the
 209 three regimes predicted by Corollary 4 (§4.2, Table 2). Together these locate useful and fail-
 210 ing regimes of the heuristic. Additional controls (E1a-Ind 120 non-principal-angle pairs, E1b
 211 30 real pairs, E1c 57 pairs, E_diverse cross-architecture, E_downstream r_{eff} /LP-accuracy)
 212 all agree with the same picture and are deferred to Appendix M. Real-data feature matri-
 213 ces are L2-row-normalized and subsampled to N_{sub} samples; effective rank uses Eq. (1);
 214 bootstrap CIs use $B = 10,000$ resamples. E1a is the explicit exception: it uses exact unnor-
 215 malized float64 matrices because row normalization would alter the proportional spectrum
 216 being checked.

217 4.1 Step 1 — Idealized-Model Agreement: Synthetic Controlled Grid (E1a, 280 pairs)

218 As a within-model sanity check, we evaluate Collapse-4S on 280 controlled synthetic
 219 pairs generated under the proportional-spectrum, paired-direction model with $\alpha \in$
 220 $\{0, 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$ and $\gamma \in \{0.1, 0.33, 0.5, 1.0, 2.0, 3.0, 10.0\}$ ($d=512$, $N=1000$,
 221 5 repeats per setting).

Table 1: E1a: Canonical Collapse-4S prediction cross-checked against direct SVD on the exact synthetic grid ($N = 280$).

Method	R^2	Max error	Superadd. (%)	$\alpha < 1$ superadd.
Canonical Collapse-4S	1.0000	1.3×10^{-13}	87.5%	100% (245/245)

Table 1 shows $R^2=1.0000$ and a maximum absolute effective-rank error of 1.3×10^{-13} against direct SVD. All 245 configurations with $\alpha < 1$ achieve 100% superadditivity; all 35 non-superadditive cases concentrate exclusively at $\alpha=1$ (identical subspaces), as expected from the generator and the boundary of Corollary 4. Visualization of the (α, γ) heatmap and superadditivity region is in Appendix M.

4.2 Steps 2–3 — Real-Data Generalization and Regime Agreement (E1d, 427 Pairs)

We next measure association beyond the synthetic derivation and test whether the predicted three-regime behavior is observable. The evaluation spans 5 encoder families (ResNet-50, ViT-B/16, CLIP ViT-B/32, DINO ViT-S/16, SBERT), 14 vision datasets (CIFAR-10/100, STL-10, SVHN, MNIST, Fashion-MNIST, BloodMNIST, DermaMNIST, PathMNIST, and 5 ImageNet semantic subsets animal/vehicle/artifact/food/scene), and 19 SBERT text corpora. Total: 427 pairs (91 ResNet-50, 55 ViT-B/16, 55 CLIP, 55 DINO, 171 SBERT).

Table 2: E1d: Collapse-4S association study (427 pairs, 5 probes). c is fitted separately per encoder family; calibrated R^2 is therefore descriptive rather than out-of-family generalization.

Probe	n	Superadd. %	Pearson r	c	R^2_{cal}
ResNet-50 (14 datasets)	91	100%	0.940	0.776	0.760
ViT-B/16 (11 datasets)	55	69.1%	0.898	0.647	0.487
CLIP ViT-B/32 (11 datasets)	55	29.1%	0.950	0.595	0.839
DINO ViT-S/16 (11 datasets)	55	36.4%	0.913	0.584	0.554
Vision (all probes)	256	64.5%	0.965	per-probe	0.942
+ SBERT text (19 corpora)	+171 (427)	78.2%	0.970	per-probe	0.952

Association (Step 2). Across the 427 evaluated pairs, the formula has overall Pearson $r=0.970$ (uncalibrated) and $R^2_{\text{cal}}=0.952$ after per-encoder-family calibration (scatter visualization in Appendix M, Fig. 5). The correlation measures ranking on this benchmark; fitted calibration does not remove the non-identifiability of Proposition 5.

Regime agreement (Step 3). (i) ResNet-50 achieves 100% superadditivity across the 91 evaluated pairs (including medical and ImageNet), consistent with the balanced- (γ, α) diagnostic of Corollary 4. (ii) Supervised/self-supervised divide emerges in the model’s directional-overlap regime: CLIP and DINO achieve only 29%–36% superadditivity, with failures concentrating entirely in cross-domain pairs (all evaluated handwritten×ImageNet and natural×digit pairs fail) while within-domain pairs remain near-perfect. A bimodal principal-angle distribution and extreme condition numbers $\kappa \in [10^4, 10^6]$ are associated with these failures (Figure 2). This pattern is consistent with the directional-collapse diagnostic of Sec. 3.3, but the comparison is not a causal identification.

Quantifying the bimodal gap. Per-probe calibrated R^2 : CLIP 0.84 > ResNet 0.76 > DINO 0.55 > ViT 0.49; ranking accuracy is uniformly high ($r \geq 0.90$). Figure 2(a,c) shows that CLIP/DINO’s two-cluster $\cos^2 \theta_i$ distribution ($V_\alpha > 0.07$) inflates the observed prediction bias, identifying a concrete departure from the global summary approximation rather than a black-box failure (full mechanism in Appendix E).

(iii) Energy collapse appears in ViT’s 17 non-superadditive pairs, all in $\gamma \notin [0.5, 2]$, matching the summary model’s energy-imbalance diagnostic. (iv) Cross-modal check: on 171 SBERT text pairs the ranking correlation drops to $r=0.793$ (98.8% superadditive), so the data do not support a modality-agnostic claim. Absolute calibration, however, requires modality-

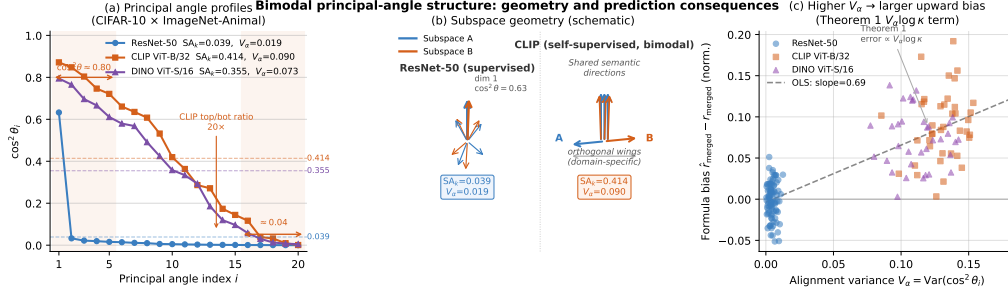


Figure 2: Bimodal principal-angle structure (CIFAR-10 $\times$ ImageNet-Animal, $k=20$). (a) Per-dimension $\cos^2 \theta_i$: ResNet-50 ($SA_k=0.039, V_\alpha=0.019$) decays monotonically from one dominant direction; CLIP ($SA_k=0.414, V_\alpha=0.090$) and DINO ($SA_k=0.355, V_\alpha=0.073$) form a two-cluster step (top-5 ratios $20\times/31\times$ over bot-5), violating the uniform- α assumption. (b) Geometric interpretation: one shared spine plus orthogonal domain-specific wings. (c) Alignment variance V_α predicts upward formula bias across 166 pairs (OLS slope 0.69), identifying alignment heterogeneity as an empirical failure indicator.

specific tuning: the vision-only fitted regression ($c=0.946-0.175 \log(N/r_{\text{eff}})$, Appendix L) explains $R^2=0.96$ of vision variance but only $R^2=0.08$ of SBERT variance, indicating that text encoders have a qualitatively different spectral bias. A text-specific calibration is left for future work.

Evaluation summary. Steps 1–3 establish within-model agreement, benchmark association, and observable qualitative regimes. They neither validate Collapse-Gram- L nor establish universal exactness: Proposition 5 proves that four summaries cannot determine merged rank in general. Cross-dataset calibration generalization, cross-probe consistency, and r_{eff}/LP -accuracy correspondence ($\rho=0.90$ within ViT family) are in Appendix M.

5 Two-Stage Data-Pool Compression

The objective of Stage 1 is a label-blind shortlist, not a final utility decision. Collapse-Gram- L applies Algorithm 1 to the composable sketches of Sec. 3.5. In the current Stage 2 audit, each candidate adapter is trained with that source’s labels; repeated cross-validation inside target-train fits ridge probes and selects the source. The target-test split is opened once after selection is frozen. This is a source-supervised adaptation protocol with target-validation selection, not a target-trained adapter. The separation is necessary because spectral diversity alone cannot determine arbitrary target utility.

Corrected protocol. Source rows for Stage 1 are sampled without labels, and unequal-size sources use the declared equal-source weighting $X_j = H_j/\sqrt{N_j}$ ($\beta_j = 1$). We compare Collapse-Gram- L against rank-only, DPP, Facility Location, agglomerative medoids, leverage-score selection, random selection, and Full-Gram greedy; an exhaustive merged-rank oracle is included only when the predeclared combination-count limit permits it. All Stage-2 candidates receive identical source-supervised optimization steps and seeds, with frozen-identity and untrained random-adapter controls. Target-train cross-validation is the only target signal used before selection and all adapter checkpoints are hashed. A selection manifest is then written and hashed before a separate held-out command is permitted to open the target-test cache. Manifests bind code, inputs, preprocessing, weights, sampled indices, checkpoints, and hyperparameters by SHA-256; incomplete source/seed groups are never resumed as completed runs. Overlap detection is reported against same-dataset matched negatives, and the 0% overlap condition has no parent ground truth. To estimate exhaustive shortlist regret, the audit trained all candidate adapters offline; its total wall time is therefore not a measured deployment saving.

Audited historical pilot. The pre-correction 21-source pilot used label-stratified source caches, queried target test before freezing selection, and did not evaluate Collapse-Gram- L .

Table 3: Corrected controlled rank sweep. Gains are relative to rank-only in exact final merged effective rank. All 960 intervals per row cover the exact selected-prefix score.

L	Gram gain	DPP gain	Mean error / bound	Certified steps
10	+2.09%	+4.90%	2.558/5.944	0/2,400
20	+3.81%	+5.48%	1.877/5.505	0/2,400
50	+4.94%	+5.68%	0.996/4.455	0/2,400

Its DPP and rank-only Recall@10 were both 85.71%, with strong near-tie sensitivity; we retain it only as a historical audit baseline and report the corrected results below.

Corrected controlled evidence. An intermediate `corrected_v2` run was invalidated because of a statistic-definition mismatch and incomplete provenance. In the replacement `corrected_v3` sweep, four encoders, ten construction seeds, six overlap levels, and four budgets give 960 configurations per rank. Table 3 reports macro-averages over encoders. Every selected-prefix interval contains the exact score, but all intervals overlap at every evaluated decision. The bound is valid yet operationally vacuous at these ranks. DPP is stronger in this controlled spectral-retention task; this is a recorded unequal-byte comparison, not a communication-matched Pareto result. A separate $L = 20$ exact-reference pilot (four encoders, three seeds, 288 configurations, Random-100) gives rank- L /DPP/Full-Gram gains of 3.65/5.37/5.81% over rank-only. Its intervals cover 288/288 prefixes, but certify 0/720 steps and match 0/288 Full-Gram ordered prefixes; Full-Gram here is a greedy reference, not a global oracle.

Corrected held-out utility evidence. The natural-pool rerun contains 21 sources, 7 targets, 4 encoders, and 5 adapter seeds. At shortlist 10/21, strict oracle recall is 75.00% for rank- L , 85.71% for rank-only and DPP, and 89.29% for Full-Gram greedy; none reaches the predeclared 90% threshold. Their held-out regrets within the adapter family are small (0.00022–0.00082), but the family itself fails the source-independent control. Mean held-out accuracy is 0.83654 for frozen identity, 0.77144 for an untrained random adapter, and 0.76951 for the validation-oracle source-supervised adapter. Oracle minus identity is -0.06703 (target-cluster 95% CI $[-0.09771, -0.04400]$); every deterministic selected adapter loses to identity in all 28 encoder-target units. Thus small regret cannot be interpreted as preserved utility or beneficial data use.

Interpretation. The theory controls spectral-score approximation only. The corrected sweep confirms interval coverage but shows that coverage alone need not produce a usable decision certificate. Stage 2 then rejects the current source-supervised adapter family before any practical-compression claim is made. Exact identity, tie-aware recall, regret, and source-independent utility controls are therefore separate quantities.

6 Conclusion

We separated exact spectral structure, bounded approximation, and task-dependent evidence. Gram additivity and the block-compatible local model are exact; the counterexample rules out universal prediction from the legacy summaries. The rank- L sketch has a valid deterministic error interval, but the corrected sweep obtains zero decision certificates for $L \leq 50$ and finds DPP at least as competitive. The held-out audit is more decisive: even the validation-oracle source-supervised adapter is substantially worse than frozen identity. The current contribution is therefore an exact local analysis, an auditable approximation protocol, and a documented negative result—not a validated practical compressor. Equal-byte Pareto evidence and a downstream adapter family that first beats identity are prerequisites for a positive application claim. Code is available at <https://anonymous.4open.science/r/collapsemodel-1DB6>.

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A Limitations

The four summaries $(r_A, r_B, \gamma, \alpha)$ are not sufficient to determine merged effective rank in general (Proposition 5). The exact 2×2 decomposition requires mode-wise pairing and decoupling; the closed entropy expression additionally requires proportional spectra and uniform alignment. We do not claim a universal error bound for Collapse-4S. The empirical calibration relation (Eq. 25) is fitted on few encoder families and transfers poorly to SBERT. The rank- L theorem requires an exact PSD truncation (or valid tail upper bounds) under fixed, subset-independent preprocessing; data-dependent global centering would break direct composability. Its intervals are too wide to certify any tested choice at $L \leq 50$, and increasing L increases communication; the current baselines are not equal-byte matched. Forming a Gram matrix also squares the condition number, so extremely ill-conditioned inputs require direct-SVD cross-checks or explicit numerical slack. Finally, unlabeled geometry cannot determine arbitrary labels. The corrected natural pool remains domain imbalanced, and the source-supervised bottleneck adapter is a narrow proxy that fails against frozen identity. LoRA and multi-pool training were consequently not run under the preregistered stopping rule. Broader domains and a useful adapter family would be required before claiming practical utility.

B Proofs of the Exact Gram Results

Proposition 1. From the compact SVDs, $H_A^\top H_A = V_A \Sigma_A^2 V_A^\top$ and $H_B^\top H_B = V_B \Sigma_B^2 V_B^\top$. Because the first columns of Q_A are V_A , $Q_A^\top V_A \Sigma_A^2 V_A^\top Q_A = \tilde{\Sigma}_A^2$. Writing $C = Q_A^\top V_B$ gives $Q_A^\top V_B \Sigma_B^2 V_B^\top Q_A = C \Sigma_B^2 C^\top$. Adding proves Eq. (4). Orthogonal similarity preserves eigenvalues, whose square roots are the merged singular values. $\square$

Lemma 2.

Under paired, decoupled directions, the i -th right singular vector of H_B is $V_{B,i} = \cos \theta_i V_{A,i} + \sin \theta_i V_{\perp,i}$, where $V_{A,i}$ is the i -th right singular vector of H_A and the two-dimensional planes are mutually orthogonal. In the basis $\{V_{A,i}, V_{\perp,i}\}$:

$$H_A^\top H_A \text{ (block)} = \sigma_{A,i}^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad (16)$$

$$H_B^\top H_B \text{ (block)} = \sigma_{B,i}^2 \begin{pmatrix} \cos^2 \theta_i & \cos \theta_i \sin \theta_i \\ \cos \theta_i \sin \theta_i & \sin^2 \theta_i \end{pmatrix}. \quad (17)$$

Define the mode-wise ratio $\gamma_i = \sigma_{B,i}/\sigma_{A,i}$; no global proportional-spectrum assumption is required. Summing gives

$$G_i = \sigma_{A,i}^2 \begin{pmatrix} 1 + \gamma_i^2 \alpha_i & \gamma_i \sqrt{\alpha_i(1 - \alpha_i)} \\ \gamma_i \sqrt{\alpha_i(1 - \alpha_i)} & \gamma_i^2(1 - \alpha_i) \end{pmatrix}, \quad (18)$$

where $\alpha_i = \cos^2 \theta_i$. Its trace is $\sigma_{A,i}^2(1 + \gamma_i^2)$ and determinant is $\sigma_{A,i}^4 \gamma_i^2(1 - \alpha_i)$, yielding Eq. (5) by the quadratic formula. $\square$

Proposition 3.

With $\gamma_i = \gamma$ and $\alpha_i = \alpha$, define $c_\pm = \sqrt{(A \pm D)/2}$. Lemma 2 gives merged singular values $\{c_+ \sigma_{A,i}\}_i \cup \{c_- \sigma_{A,i}\}_i$. After nuclear-norm normalization their probabilities are qp_i^A and $(1 - q)p_i^A$, where $q = c_+/(c_+ + c_-)$. Hence

$$H_{\text{merged}} = H_b(q) + qH(p^A) + (1 - q)H(p^A) \quad (19)$$

$$= H_b(q) + \log r_A. \quad (20)$$

Exponentiating proves Eq. (7). Proportional spectra have identical normalized singular-value distributions, so $r_A = r_B$. $\square$

Theorem 1. Write $E_S = \sum_{j \in S} E_j$. For PSD matrices, $\text{tr} \sqrt{A + B} \leq \text{tr} \sqrt{A} + \text{tr} \sqrt{B}$: apply the nuclear-norm triangle inequality to $[A^{1/2}, B^{1/2}]$. Hence

$$\text{tr} \sqrt{E_S} \leq \sum_j \tau_j, \quad \text{tr} \sqrt{E_S} \leq \sqrt{\text{rank}(E_S) \text{tr}(E_S)} \leq \sqrt{\min(d, \sum_j u_j) \sum_j \omega_j}, \quad (21)$$

so $\text{tr} \sqrt{E_S} \leq T_S$. Let $a_i = \sqrt{\lambda_i(\tilde{G}_S)}$ and $b_i = \sqrt{\lambda_i(G_S)}$ in decreasing order. Since $G_S \succeq \tilde{G}_S$, eigenvalue monotonicity gives $b_i \geq a_i$. With $\tilde{\nu} = \sum_i a_i$ and $\delta = \sum_i (b_i - a_i)$, trace subadditivity gives $0 \leq \delta \leq T_S$. If $\delta > 0$, the normalized true spectrum is

$$p = \frac{\tilde{\nu}}{\tilde{\nu} + \delta} \tilde{p} + \frac{\delta}{\tilde{\nu} + \delta} q, \quad q_i = \frac{b_i - a_i}{\delta}. \quad (22)$$

Thus $\text{TV}(p, \tilde{p}) \leq \delta/(\tilde{\nu} + \delta) \leq \varepsilon_S$. Fannes–Audenaert entropy continuity on a support of size at most D_S gives Eq. (14). The $\delta = 0$ and $D_S = 1$ cases are immediate. $\square$

Corollary 6. The theorem places every exact candidate score in $[\hat{F}_j - e_j, \hat{F}_j + e_j]$. Equation (15) puts the lower endpoint of j^* strictly above every competing upper endpoint. $\square$

Status of Eq. (8). For unequal or non-proportional spectra, the two branches need not inherit the input distributions, and q can vary by mode. Equation (8) therefore defines a structured approximation; no exactness or universal error bound is asserted for it.

C Optimization Limits

Effective-rank maximization is not generally monotone. Take $G_A = \text{diag}(1/2, 1/2)$ and $G_B = \text{diag}(1, 0)$. Then $\mathcal{R}(G_A) = 2$, whereas $\mathcal{R}(G_A + G_B) = 1.928623 < 2$. It is also not generally submodular. Let $G_0 = \text{diag}(1/5, 4/5)$, $G_1 = \text{diag}(0, 1)$, and $G_2 = \text{diag}(1, 0)$. For $A = \{0\} \subset B = \{0, 1\}$, direct evaluation of $F(S) = \log \mathcal{R}(\sum_{j \in S} G_j)$ gives

$$F(A \cup \{2\}) - F(A) = 0.051522 < 0.125701 = F(B \cup \{2\}) - F(B), \quad (23)$$

violating diminishing returns. Thus standard monotone-submodular greedy guarantees do not apply; exhaustive search is used only as a finite small-pool reference.

D Bootstrap Confidence Intervals

Table 4: Bootstrap 95% CIs ($B = 10,000$, seed=42) for all key metrics.

Experiment	Metric	Point estimate	95% CI	SE
E1a (280 pairs)	R^2	1.0000	[1.0000, 1.0000]	0.000000
E1a	Superadd. ($\alpha < 1$)	1.0000	[1.000, 1.000]	0.000000
E1b (30 pairs)	Pearson r	0.9478	[0.906, 0.978]	0.019
E1b	Superadd. %	0.8667	[0.733, 0.967]	0.062
E1c (57 pairs)	R^2 (global)	0.049	[-0.346, 0.287]	0.161
E1c	Pearson r (global)	0.928	[0.894, 0.949]	0.014
E1c	Superadd. %	0.930	[0.860, 0.982]	0.034
E1c (ResNet-50)	Superadd. %	1.000	[1.000, 1.000]	0.000
E1c (ResNet-50)	Pearson r	0.934	[0.885, 0.965]	0.021
E1c (ViT-B/16)	Superadd. %	0.733	[0.467, 0.933]	0.114
E1c (ViT-B/16)	Pearson r	0.945	[0.883, 0.982]	0.026
E1c (SBERT)	Superadd. %	1.000	[1.000, 1.000]	0.000

E Bimodal Principal-Angle Structure in Self-Supervised Encoders

The per-dimension $\cos^2 \theta_i$ profiles and their prediction consequences are shown in Figure 2 (main text): CLIP/DINO exhibit a two-cluster step structure (top-5 ratios $20 \times / 31 \times$ over bot-5, $V_\alpha = 0.090/0.073$), while ResNet-50 decays monotonically ($V_\alpha = 0.019$).

Two distinct mechanisms drive the low- R^2 probes. For DINO ($R^2 = 0.55$), the cause is the bimodal $\cos^2 \theta_i$ analysed below. For ViT ($R^2 = 0.49$, supervised), the cause is energy imbalance rather than bimodality: 17/17 ViT non-superadditive pairs lie in $\gamma \notin [0.5, 2]$ (cross-resolution pairs such as MNIST $\times$ ImageNet), where the global energy summary is extreme

and the two-component mixture approximation degrades. ViT’s principal-angle distribution remains near-uniform (no bimodality); its lower R^2 instead empirically marks the energy-imbalance regime as a reliability boundary. We retain the full ViT row in Table 2 for transparency, flagging $\gamma \in [0.5, 2]$ as ViT’s reliable applicability window.

Honest scope. We treat the bimodal failure mode as a genuine boundary of the current formula, not a feature. The uniform- α assumption underlying The global-summary approximation is stressed whenever the principal-angle distribution is bimodal, and self-supervised contrastive/distillation encoders (CLIP, DINO) systematically produce such distributions on cross-domain pairs. Within this regime, the calibrated formula still ranks correctly (Pearson $r \geq 0.90$ on all four probes, Table 2) but the absolute prediction $\hat{r}$ is biased upward and the superadditivity-decision boundary is shifted, explaining the 29–36% observed superadditivity rates on CLIP/DINO cross-domain pairs. What the Collapse Model does contribute here is mechanistic localisation: the bimodal top/bottom ratio $>20\times$ in $\cos^2 \theta_i$ (Figure 2) exposes alignment heterogeneity discarded by the scalar summary, rather than leaving the failure as a black-box surprise.

What does not work. A naive energy-weighted patch $\bar{\alpha}_w = \sum_i (\sigma_i / \|H\|_*) \cos^2 \theta_i$ worsens predictions (Pearson r drops 1–2%, MAE rises 6–8%), because it overweights the most-aligned direction’s redundancy while suppressing intermediate-direction contributions.

What we believe will work, and how to test it. The natural next step is a piecewise paired-direction model: partition the k directions into a high-alignment cluster ($\cos^2 \theta_i \geq \tau$) and a low-alignment cluster, apply Lemma 2 separately within each cluster, and combine the resulting branch spectra. This retains more of the distribution of α_i than a single mean, but it still requires controlling spectral pairing and cross-direction coupling. We leave this extension and its error analysis to future work.

F Historical Compression Pilot at $M=14$ Vision and $M=7$ Text Pools

This pre-audit experiment applies Collapse-4S greedy across three encoder families and two modalities, using FacLoc / k -Medoids as baselines. It predates the label-blind cache, source weighting, and Collapse-Gram- L implementation; all numbers below are descriptive historical results.

Table 5: Diversity retention $r_{\text{eff}}^{\text{sel}}/r_{\text{eff}}^{\text{full}}$ (%) across strategies, architectures, and modalities. FacLoc = Facility Location (submodular greedy on SA_k kernel); kMedoids = PAM on $1-\text{SA}_k$ distance. Random is the 10-seed mean $\pm$ std. The table evaluates spectral retention, not a task-utility guarantee.

Probe	M	k (ratio)	Collapse-4S	FacLoc	kMedoids	Random	LP _{4S}
Vision							
ResNet-50	14	3 (4.7 $\times$)	101.4	87.8	87.5	84.3 $\pm$ 10.0	0.754
ResNet-50	14	5 (2.8 $\times$)	105.7	95.6	92.6	88.7 $\pm$ 5.3	0.754
ResNet-50	14	7 (2.0 $\times$)	105.9	98.4	98.0	93.1 $\pm$ 7.6	0.747
ViT-B/16	11	3 (3.7 $\times$)	103.6	—	—	92.0 $\pm$ 3.3	0.779
ViT-B/16	11	5 (2.2 $\times$)	104.9	—	—	94.1 $\pm$ 3.7	0.779
Text							
BERT L6	7	3 (2.3 $\times$)	103.6	—	—	97.8 $\pm$ 5.0	—
RoBERTa L6	7	3 (2.3 $\times$)	103.8	—	—	97.0 $\pm$ 6.1	—
GPT-2 L6	7	3 (2.3 $\times$)	104.8	—	—	92.8 $\pm$ 9.5	—
GPT-2 L6	7	5 (1.4 $\times$)	106.0	—	—	96.7 $\pm$ 4.7	—

On this ResNet-50 pool, Collapse-4S has 7.5–13.6 percentage points higher spectral retention than Facility Location and 7.9–13.9 points higher than k -Medoids. The accompanying LP values are a narrow pilot and were not produced by the corrected two-stage protocol; they do not establish that the retention ordering transfers to new targets.

525 G r_A+r_B Heuristic: Safe Regime and Failure Mode

526 When does r_A+r_B suffice, and when does it fail? On clean curated pools, the r_A+r_B
527 and Max- r_{eff} heuristics often select the same subset as Collapse-4S (within 0.7pp in the
528 historical $M=43$ stress test). This agreement is consistent with the summary diagnos-
529 tic in Corollary 4: when all top- r_{eff} candidates have low mutual alignment ($\bar{\alpha}<0.2$, i.e.
530 rank-ratio $r_{\text{dom}}/r_{\text{sub}}<\rho_c$), the superadditivity bonus $H_b(q)$ is dominated by the rank term
531 $q \log r_{\text{dom}} + (1-q) \log r_{\text{sub}}$, making individual rank a near-sufficient statistic. Collapse-4S
532 thus serves a different role than “beating r_A+r_B everywhere”: (i) A priori low-redundancy
533 diagnostic. $\rho_c(\bar{\alpha}, \bar{\gamma})$ computed from per-dataset summaries indicates when r_A+r_B is likely to
534 agree without actually running competing strategies. On our ResNet-50 vision pools, $\bar{\alpha}<0.2$
535 between top- r_{eff} pairs places $r_{\text{dom}}/r_{\text{sub}}$ well inside the predicted agreement regime ($\rho_c>4$ at
536 $\bar{\alpha}\leq 0.2$, Cor. 4), matching the observed heuristic-equivalence at $M=14$ and $M=43$. (ii) De-
537 tection of the unsafe regime. When candidates contain near-duplicate or strongly aligned
538 subspaces, the α channel supplies information that rank-only omits. The clone experiment
539 below (r_A+r_B retention drops 25.5%, Collapse holds) demonstrates this on a controlled
540 stress test; the naturally bimodal CLIP/DINO cross-domain pairs (Sec. 4.2, $\alpha \in [0.35, 0.41]$
541 with top-5 $\cos^2 \theta \geq 0.7$) are real-world settings where the α signal is non-negligible and
542 the rank-only heuristic can fail. (iii) Diagnosis without execution. The Collapse-4S regime
543 classification is computable from (γ, α) , but Proposition 5 makes it a heuristic diagnostic,
544 not a certified decision rule.

545 Redundancy clone trap: failure mode of r_A+r_B . To stress-test the alignment term un-
546 der constructed redundancy, we construct a clone attack scenario: the highest- r_{eff} dataset
547 in the pool (STL-10, $r_{\text{eff}}=994$) is cloned via random rotation $H_{\text{clone}}=H_{\text{orig}}Q(\theta)$ ($\theta \in$
548 $\{0.01, 0.03, 0.05, 0.07, 0.10\}$ radians), producing 5 clones that preserve the original r_{eff} but
549 are highly aligned with the original ($\text{SA}_k>0.99$). On a pool of 5 real + 5 cloned ResNet-50
550 datasets, r_A+r_B greedy at $k=3$ selects 2/3 clones ($r_{\text{eff}}^{\text{merged}}=997$), while Collapse-4S assigns
551 a redundancy penalty via high α , selecting 0/3 clones ($r_{\text{eff}}^{\text{merged}}=1251$, +25.5%). At $k=5$
552 the gap persists: r_A+r_B selects 4/5 clones ($r_{\text{eff}}=998$), Collapse only 2/5 ($r_{\text{eff}}=1233$). This
553 result is consistent across all 5 θ values ($\theta=0.01\text{--}0.10$, SA_k from 0.9999 to 0.9891). Because
554 the duplicated source was deliberately chosen from the highest-rank candidate, this is an
555 adversarial stress test rather than evidence about natural-pool frequency. It shows only
556 that alignment can help in this constructed case.

557 H P2-E: Hybrid Two-Signal Strategy

558 The historical $M=40$ pilot suggested a two-signal hybrid: seed $\lceil k/2 \rceil$ datasets via one signal,
559 fill the remaining $\lfloor k/2 \rfloor$ via the other. We test three orderings on that same $M=40$ ResNet-
560 50 pool: Col→Cen (Collapse first, Centroid-FF on residuals), Cen→Col (Centroid-FF first,
561 Collapse on residuals), and Col→Dom (Collapse first, DomainDiverse on residuals).

Table 6: Two-signal hybrid strategies on $M=40$ ResNet-50. Pure-strategy reference rows are shown for comparison. Bold marks the best LP at each k .

Strategy	$k=5$ (8.0×)		$k=7$ (5.7×)		$k=10$ (4.0×)	
	Ret%	LP	Ret%	LP	Ret%	LP
Pure (reference)						
Collapse	117.2	0.7576	117.4	0.7578	116.2	0.7576
Centroid-FF	93.4	0.7570	96.7	0.7624	100.5	0.7630
DomainDiverse	114.7	0.7634	111.8	0.7596	113.8	0.7544
Hybrid two-stage						
Col→Cen	110.2	0.7534	111.9	0.7496	113.8	0.7546
Cen→Col	106.7	0.7596	113.6	0.7610	115.0	0.7602
Col→Dom	114.7	0.7634	116.0	0.7562	113.8	0.7544

Historical result. At $k \in \{7, 10\}$ (the operational compression range), Cen $\rightarrow$ Col achieves LP 0.7610/0.7602, exceeding pure Collapse (0.7578/0.7576) by +0.0032/+0.0026 while sacrificing only 3.8/1.2 pp of retention. At $k=10$, Cen $\rightarrow$ Col trades 1.2 percentage points of retention for 0.0026 higher LP; it does not dominate pure Collapse on both metrics. Compared to pure Centroid-FF, Cen $\rightarrow$ Col recovers 13.3 pp of retention at $k=5$, 16.9 pp at $k=7$, and 14.5 pp at $k=10$ while keeping LP within ± 0.0028 . Col $\rightarrow$ Cen, in contrast, hurts both metrics: once Collapse has selected high- r_{eff} datasets, adding "different prototype" datasets dilutes the spectrum without recovering centroid coverage. Col $\rightarrow$ Dom reduces to DomainDiverse on this pool because DomainDiverse already picks STL-10 and fashion_mnist (the highest- r_{eff} datasets in their respective domains) that Collapse would also select.

Implication. The ordering matters: starting with Centroid-FF establishes a geometric backbone that Collapse then augments with high- r_{eff} complementary directions. The summary formula's role in this hybrid is to fill the residual budget with diversity-maximizing choices, exactly the regime where it operates with summary statistics alone (the centroid-FF prefix can be precomputed once from already-shared centroids, no feature-matrix exchange needed). This single-pool result motivates, but does not validate, a composable two-signal strategy. It must be repeated with the corrected protocol and independent targets.

I E5 Supplementary: Per-Target Selection Details

Per-target regret table ($M=10$ pool).

Strategy	Mean regret (%) / Mean recall (%)			
	$K=1$	$K=2$	$K=3$	$K=5$
Collapse-4S	0.0 / 100	0.0 / 100	0.0 / 100	1.0 / 87
Max- r_{eff}	0.0 / 67	2.0 / 67	0.9 / 89	1.0 / 87
Δ_{dir} -greedy	20.3 / 0	16.7 / 17	17.0 / 33	13.3 / 40
Random	15.2 / -	17.5 / -	16.1 / -	11.1 / -

For BloodMNIST ($K=1$), Max- r_{eff} selects STL-10 (highest individual $r_{\text{eff}}=357$) but Collapse-4S selects CIFAR-100 ($r_{\text{eff}}=356$, marginally lower) because the latter has lower alignment with BloodMNIST ($\alpha=0.18$ vs. 0.21), yielding a higher predicted merged rank—confirmed by $r_{\text{eff}}^{\text{merged}} = 506.8$ vs. 506.4.

r_A+r_B greedy baseline. We evaluate an r_A+r_B greedy strategy that selects the candidate with the highest individual r_{eff} at each round. On the $M=10$ pool, this baseline matches Collapse-4S at $K \leq 3$ (identical subset, 0% regret), diverging only at $K=5$ where Collapse-4S retains 100% recall while r_A+r_B drops to 87%. This mirrors the ablation finding: r_A+r_B suffices when candidates have distinct individual ranks, but fails when alignment discrimination (α) is needed.

Downstream LP validation. We train a linear probe on the merged features from each strategy's K -subset and evaluate on held-out test data. All four strategies (Exhaustive, Collapse-4S, r_A+r_B , Max- r_{eff}) produce identical LP accuracy across all 6 tested configurations (CIFAR-10 and STL-10 targets, $K \in \{1, 2, 3\}$), ranging from 0.776 (CIFAR-10, $K=3$) to 0.979 (STL-10, $K=1$). Because every method has identical LP accuracy here, this experiment does not distinguish selectors and cannot show that maximizing merged r_{eff} improves downstream utility.

J E7 Supplementary: kNN Downstream Validation

The main text uses linear-probe (LP) accuracy as one downstream metric. Here we compare whether k -nearest-neighbour (kNN) gives similar rankings on the evaluated configurations; this cannot establish LP as sufficient for arbitrary downstream tasks.

Part B: LP-kNN rank correlation. We compute LP, kNN-5, and kNN-20 accuracy on 5 single datasets (CIFAR-10, CIFAR-100, STL-10, SVHN, Fashion-MNIST) across 11 encoder families (ResNet-50/101, ViT-B/16, DeiT-B, Swin-Tiny, BEiT-Base, ConvNeXt-Tiny,

606 DenseNet-121, EfficientNet-B4, MaxViT-Tiny, RegNet-Y), yielding 53 observations. Spear-
 607 man rank correlation: $\rho(\text{LP}, \text{kNN-5})=0.980$, $\rho(\text{LP}, \text{kNN-20})=0.973$ —LP and kNN rankings
 608 are near-identical.

609 Part A: compression advantage under kNN. Using the $M=6$ ResNet-50 pool from Sec. 5, we
 610 evaluate Collapse Greedy vs. 10 random seeds using LP, kNN-5, and kNN-20 on the merged
 611 features (averaged over 6 targets).

k	Strategy	LP	kNN-5	kNN-20
3	Collapse	73.5	62.4	61.2
	Random (μ)	72.6	58.2	53.8
	Δ	+0.9	+4.2	+7.4
5	Collapse	71.5	57.8	53.5
	Random (μ)	71.1	56.8	51.1
	Δ	+0.4	+1.0	+2.4

613 On this historical $M=6$ pool, the Collapse-4S-random difference is larger for kNN than
 614 for LP; at $k=3$, for example, it is 7.4 percentage points for kNN-20 and 0.9 for LP. The
 615 high single-dataset LP-kNN correlation ($\rho>0.97$) and the strategy comparison reuse a small,
 616 shared pool, so we treat LP only as one proxy rather than a sufficient one.

617 K Additional Notes

618 Note on multiple comparisons. The within-probe bootstrap analyses (Sec. 5) involve 11
 619 probes tested independently. Since the primary claim is about the average $\Delta\rho$ across
 620 probes (not individual-probe significance), Bonferroni correction is not required for the
 621 aggregate Fisher z -test ($p=0.74$). For the vision-only bootstrap ($\Delta\rho=+0.005$, 95% CI
 622 $[+0.001, +0.011]$), a conservative Bonferroni threshold of $\alpha=0.05/11=0.0045$ would still be
 623 met ($P(\text{Collapse} > r_{\text{sum}})=0.99$).

624 L Scope of the Paired-Direction Model

625 Remark 1 (Exact and approximate layers). Proposition 1 is unconditional and retains the
 626 full coupling matrix $C = Q_A^\top V_B$. Lemma 2 is exact only when the singular directions can be
 627 paired into mutually orthogonal two-dimensional planes. Proposition 3 additionally requires
 628 mode-wise proportional spectra and uniform alignment; these conditions imply $r_A = r_B$.
 629 The practical predictor Eq. (8) allows $r_A \neq r_B$ by interpolating two input entropies, and is
 630 therefore a modeling approximation rather than a consequence of the proportional-spectrum
 631 proposition.

632 Remark 2 (Information discarded by global summaries). The global ratio $\gamma = \|H_B\|_* / \|H_A\|_*$
 633 does not imply $\sigma_{B,i} = \gamma\sigma_{A,i}$. Likewise, $\alpha = \text{SA}_k$ does not retain the distribution of α_i , its
 634 association with spectral mass, or dense cross-direction coupling. Proposition 5 gives an
 635 explicit pair of examples with identical $(r_A, r_B, \gamma, \alpha)$ and different merged effective ranks.
 636 Consequently no estimator based only on these four scalars can be universally exact.

637 Remark 3 (Empirical generalization). The summaries in Eq. (8) remain well-defined out-
 638 side the paired-direction model. Their empirical correlations on synthetic out-of-model
 639 generators and real encoders quantify predictive utility, not theorem validity. Alignment
 640 heterogeneity, non-proportional spectra, and cross-direction coupling are observable failure
 641 sources; calibration corrects average scale on the evaluated encoder family but does not
 642 create a general guarantee.

643 Remark 4 (Calibrated Collapse Model). On real data the paired-direction summary model is
 644 approximate—cross-pair coupling, non-proportional spectra ($\sigma_{B,i} \neq \gamma\sigma_{A,i}$, hence $r_A \neq r_B$),
 645 and higher-order interactions introduce a systematic upward bias on the evaluated pairs.
 646 Empirically the over-prediction ratio is $1.24\text{--}1.84\times$ (E1c), and its gap is positively associated
 647 with alignment heterogeneity. Since the log-space bias has low variance (std = 0.164 vs.

mean = 0.293), a single multiplicative calibration restores quantitative accuracy:

$$\hat{r}_{\text{cal}} = c \cdot \hat{r}_{\text{merged}}, \quad c = \exp\left(-\frac{1}{n} \sum_{i=1}^n \log \frac{\hat{r}_i}{r_i}\right). \quad (24)$$

This calibration compensates for the entropy over-estimation caused by structure that the four global summaries ignore. With $c=0.746$ estimated from E1c ($n=57$), LOO-CV yields $R^2=0.855$, RMSE=87.7 (vs. uncalibrated $R^2=0.049$, RMSE=224; Sec. M.7). The constant generalizes across datasets within each encoder family: calibrating on the 30 E1b pairs ($c=0.768$) and testing on the held-out 57 E1c pairs yields $R^2=0.859$, RMSE=86.3 (Table 11). However, c varies across encoder families ($c=0.78$ for ResNet-50 vs. $c=0.65$ for ViT-B/16), reflecting systematic differences in how each architecture’s spectral structure deviates from the paired-direction summary model. On the evaluated encoder families, this variation is fit by:

$$c = 0.946 - 0.175 \log(N/r_{\text{eff}}), \quad R^2=0.963, \quad \text{LOO mean error}=0.027, \quad (25)$$

This small-sample regression is descriptive and requires validation on a new encoder family before use; its in-sample R^2 is not evidence of a universal law.

L.1 Why a Task-Aware Second Stage Is Necessary

Unlabeled Grams contain no information about an arbitrary target labeling. Indeed, two tasks may share exactly the same frozen features while assigning labels to different directions, reversing the utility order of two source subsets. Therefore no label-free spectral score can universally guarantee downstream utility. Our Stage-2 validation is not a consequence of Theorem 1: it is the separate empirical step needed to measure target-conditioned utility under equal training compute.

Algorithm 1 Collapse-Gram- L Greedy Screening

Require: Summaries $\{(B_j, \tau_j, \omega_j, u_j)\}_{j=1}^M$, budget K
 Ensure: Ordered shortlist S , with a certificate flag for each step
 1: $S \leftarrow \emptyset$; $C \leftarrow \{1, \dots, M\}$
 2: for $t = 1, \dots, K$ do
 3: for each $j \in C$ do
 4: compute $\hat{F}_j$ and e_j for $S \cup \{j\}$ using Eqs. (11)–(14)
 5: end for
 6: $j^* \leftarrow \arg \max_{j \in C} \hat{F}_j$
 7: $z_t \leftarrow [\hat{F}_{j^*} - e_{j^*} > \max_{j \neq j^*} (\hat{F}_j + e_j)]$
 8: $S \leftarrow S \parallel j^*$; $C \leftarrow C \setminus \{j^*\}$
 9: end for
 10: return $(S, z_1, \dots, z_K)$

Certified Stage-1 pseudocode.

M Additional Experiments (E1a, E1a-Ind, E1b, E1c, Calibration, E_diverse, E4, E_downstream)

M.1 Ranking Ablation Table

M.2 E1a: Synthetic Controlled Validation

The E1a synthetic validation results (Table 1) are presented in the main text; Figure 3 visualizes the (α, γ) landscape and superadditivity region. Additional design details: feature pairs are generated from the proportional-spectrum paired-direction model with $\alpha \in \{0, 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0\}$ and $\gamma \in \{0.1, 0.33, 0.5, 1.0, 2.0, 3.0, 10.0\}$, $d = 512$, $N = 1000$, 5 independent repeats per setting (280 pairs). The exact float64 construction is not row-normalized; both direct SVD and the predictor use the common numerical-rank

Table 7: Ranking ablation: Spearman ρ between predictor and true merged r_{eff} (256 vision pairs, 4 encoder families).

Predictor	Channels used	ResNet-50 ($n=91$)	ViT-B/16 ($n=55$)	All vision ($n=256$)
Collapse Model	$\Sigma+V+U$	0.896	0.923	0.966
$r_A + r_B$	U only	0.892	0.946	0.963
$\max(r_A, r_B)$	U only	0.898	0.779	0.953
$1 - \alpha$ (V only)	V only	-0.289	-0.192	0.450
γ (Σ only)	Σ only	0.277	0.001	0.271

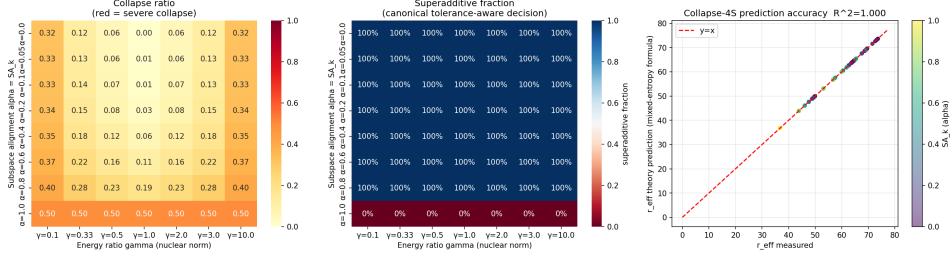


Figure 3: El1a: Collapse Model prediction accuracy on the synthetic (α, γ) grid. Left: collapse ratio heatmap. Center: tolerance-aware superadditivity region. Right: predicted vs. actual r_{eff} scatter ($R^2=1.0000$).

678 cutoff stated in Sec. 2. All 35 non-superadditive cases concentrate exclusively at $\alpha = 1$
679 (identical subspaces), confirming the degenerate boundary of Corollary 4.

680 M.3 El1a-Ind: Independent Validation on Non-Principal-Angle Matrices

681 Motivation and design. A critical concern with El1a is circularity: synthetic data gener-
682 ated from the proportional-spectrum paired-direction model satisfies the exact model as-
683 sumptions, making $R^2 = 1.0000$ a near-tautological result. To independently test whether
684 Eq. (8) remains predictive beyond its derivation assumptions, we generate 120 synthetic ma-
685 trix pairs using four generation methods that do not use that paired model: (M1) low-rank
686 Gaussian: $H = U_k \Sigma V_k^\top + \epsilon$ with random orthogonal bases; (M2) shared+private subspace:
687 $H = H_{\text{shared}} + H_{\text{private}}$ with partial overlap enforced by construction; (M3) sparse activa-
688 tions: rows drawn from k -sparse Gaussian mixtures; (M4) block-diagonal: features from
689 two independent subspaces with no principal-angle structure. $d = 512$, $N \in \{500, 1000\}$; 30
690 pairs per method.

691 Results. Table 8 shows Pearson $r = 0.9987$ overall: the formula correctly ranks all 120 pairs
692 across all four generation methods. R^2 is method-dependent: ≥ 0.99 for Gaussian-structure
693 methods (M1, M2), but strongly negative for sparse and block-diagonal structures (M3, M4),
694 because those structures systematically reduce actual r_{eff} relative to the formula’s prediction.
695 This reveals excellent ordering but poor absolute prediction on sparse and block-diagonal
696 structures, exposing structural sparsity as an empirical failure mode. Superadditivity is
697 predicted at 100% for all methods.

698 Remark 5 (Formula as directional oracle, not absolute predictor). The contrast between
699 $R^2 = -1.09$ (overall, El1a-Ind) and Pearson $r = 0.9987$ (same data) is not contradictory: R^2
700 measures absolute calibration, while Pearson r measures directional ranking. This distinc-
701 tion is crucial for the real-data results (El1c): the formula yields $R^2 \approx 0.05$ (poor absolute
702 calibration, 1.24–1.84 $\times$ over-prediction) but Pearson $r = 0.928$ (correct directional ranking).
703 For practical applications—ranking candidate merges by diversity gain, dataset selection,
704 data valuation—ranking is the operationally relevant metric, not absolute r_{eff} prediction.
705 Absolute calibration requires a correction factor (calibrated $\hat{r} = 0.81 \cdot \hat{r}_{\text{pred}}$, $R^2_{\text{cal}} = 0.862$ on
706 El1c data), which we report as a practical recommendation.

Table 8: E1a-Ind: Independent validation on 120 non-principal-angle synthetic pairs. Pearson r measures directional ranking; $R^2 < 0$ indicates systematic over-prediction on the observed generator. SA% = fraction of pairs where the formula predicts superadditivity.

Generation Method	n pairs	Pearson r	R^2	SA%
M1: Low-rank Gaussian	30	0.9985	+0.9949	100%
M2: Shared+private subspace	30	0.9982	+0.9963	100%
M3: Sparse activations	30	0.9991	−14.65	100%
M4: Block-diagonal	30	0.9996	−12.32	100%
Overall	120	0.9987	−1.09	100%

M.4 E1b: Real Dataset Pairs (30 Pairs)

Design. We evaluate all $\binom{6}{2} = 15$ pairs from {CIFAR-10, STL-10, SVHN, MNIST, Fashion-MNIST, CIFAR-100} using 2 pretrained probes (ResNet-50 [He et al., 2016] $d=2048$ and ViT-B/16 [Dosovitskiy et al., 2021] $d=768$), $N_{\text{sub}} = 500$ samples per dataset: 30 total configurations. The Collapse Model is applied with per-probe nuclear-norm γ .

Results. 86.7% of pairs are superadditive (26/30; 95% CI [73%, 97%]). Pearson $r = 0.948$ (95% CI [0.906, 0.978]) between predicted and actual merged rank. Per-probe: ResNet-50 achieves 100% superadditivity (15/15, Pearson $r = 0.968$); ViT-B/16 achieves 73.3% (11/15, Pearson $r = 0.945$).

All 4 exceptions are ViT-B/16 pairs involving CIFAR-100. The exceptions are associated with energy imbalance, consistent with the superadditivity diagnostic (Corollary 4). CIFAR-100 has 100 classes and correspondingly high intrinsic complexity: $r_{\text{eff}}(\text{CIFAR-100}) \approx 526$ under ViT-B/16, far exceeding partner datasets such as MNIST ($r_{\text{eff}} \approx 140$) and SVHN (≈ 243). This yields energy ratios $\gamma \in [2.0, 2.5]$, placing these pairs near the superadditivity boundary $\rho_c(\alpha, \gamma)$. Specifically, the dominant component absorbs the subordinate: $q \rightarrow 1$, $H_b(q) \rightarrow 0$, and $r_{\text{merged}} \rightarrow r_{\text{dom}}$, so the observed r_{merged} can fall slightly below r_{dom} due to spectral averaging. We verify this by sweeping $N_{\text{sub}} \in \{500, 1000, 2000, 4000\}$: at $N=500$, CIFAR-100’s r_{eff} is compressed to ≈ 343 (comparable to partners), giving $\gamma \approx 1$ and 100% superadditivity (5/5); as N grows to 2000+, r_{eff} stabilizes at 526, γ increases, and exceptions emerge. These quantities co-vary and are consistent with the energy-imbalance explanation, but the sweep does not identify it causally. ResNet-50 ($d=2048$, $r_{\text{eff}} \approx 988$ for CIFAR-100) yields even larger γ ratios, yet remains 100% superadditive (36/36) because its partner datasets also have higher r_{eff} in 2048-dimensional space, keeping $r_{\text{dom}}/r_{\text{sub}}$ below ρ_c .

M.5 E1c: Large-Scale Multi-Domain Validation (57 Pairs)

Design. To test generalization beyond vision pairs, we extend to three probes and three domain types: ResNet-50 on 9 datasets (36 pairs, natural+medical), ViT-B/16 on 6 datasets (15 pairs, natural), and SBERT [Reimers and Gurevych, 2019] on 4 text domains (6 pairs, text). Total: 57 pairs, 3 probes, 3 domain types. $N_{\text{sub}} = 2000$ for natural/medical; $N_{\text{sub}} = 1000$ for text.

Results. Table 9 shows strong and consistent superadditivity across all probes and domains.

Table 9: E1c: Historical Collapse-4S association study. Superadditivity rate and Pearson r (predicted vs. actual merged r_{eff}). Bootstrap 95% CI ($B = 10,000$). The formula over-predicts on all probes in this experiment; “Bias” reports the observed multiplicative ratio.

Probe / Domain	n	Superadd. % (95% CI)	Pearson r (95% CI)	Bias
ResNet-50 (natural+medical)	36	100% [100,100]	0.934 [0.885,0.965]	$1.24\times$
ViT-B/16 (natural)	15	73.3% [46.7,93.3]	0.945 [0.883,0.982]	$1.45\times$
SBERT (text)	6	100% [100,100]	0.780	$1.84\times$
Overall	57	93.0% [86.0,98.2]	0.928 [0.894,0.949]	$1.37\times$

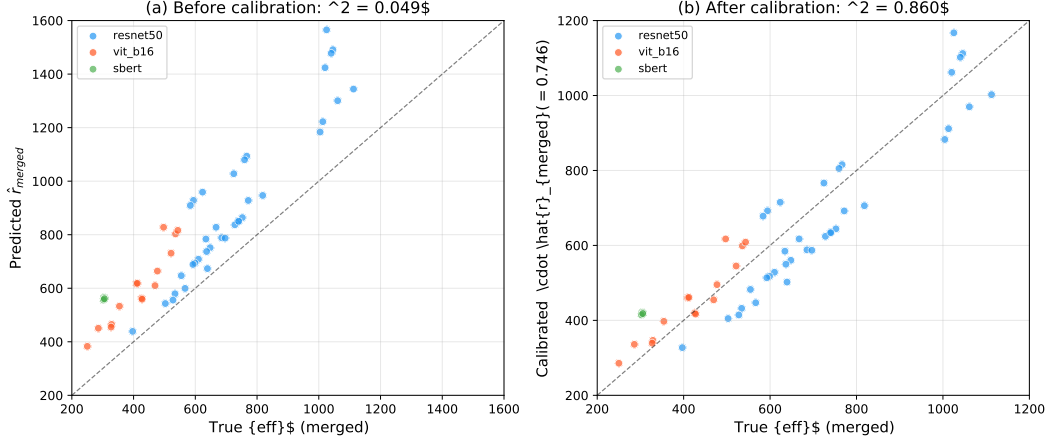


Figure 4: E1c: Predicted vs. actual merged r_{eff} for 57 real-data pairs (5 probes, 3 domain types). (a) Before calibration: systematic over-prediction ($R^2=0.049$). (b) After single-parameter calibration ($c=0.746$): $R^2=0.860$. Pearson $r=0.928$ is identical in both panels—calibration corrects scale, not ranking.

Key findings. (i) 93% superadditivity on this mixed benchmark is descriptive; the six SBERT pairs are too few to establish domain- or modality-agnostic behavior. (ii) All 4 exceptions are ViT-B/16 pairs involving CIFAR-100, attributable to energy imbalance (same mechanism as E1b, Sec. M.4): CIFAR-100’s high intrinsic complexity ($r_{\text{eff}} \approx 526$, 100 classes) creates $\gamma \in [2.0, 2.5]$, placing these pairs near the superadditivity boundary ρ_c of Corollary 4. (iii) Pearson $r = 0.928$ vs. $R^2 \approx 0.05$: these metrics measure different properties. The formula over-predicts by $1.24\text{--}1.84\times$ on these probes, so R^2 is low before calibration; Pearson r measures ordering. A single-parameter calibration (Eq. 24) restores quantitative accuracy: $R^2_{\text{cal}}=0.855$ (LOO-CV), and cross-validation from E1b to E1c yields $R^2=0.859$ (see Sec. M.7 and Remark 4). (iv) ResNet-50: 100% superadditivity (36/36) is strong empirical agreement with the summary diagnostic on this observed set. (v) Over-prediction scales with n/r_{eff} : ResNet-50 ($n = 2000 \gg r_{\text{eff}} \approx 350$, $1.24\times$) < ViT-B/16 ($n = 2000 \approx 3r_{\text{eff}} \approx 650$, $1.45\times$) < SBERT ($n = 1000 \approx r_{\text{eff}} \approx 700$, $1.84\times$). This is an observed trend in a small set, not a finite-sample theorem.

M.6 E1d: Large-Scale Validation (427 Pairs)

Design. To test the Collapse Model at scale and on ImageNet-derived features, we construct a comprehensive evaluation spanning 5 encoder families (ResNet-50, ViT-B/16, CLIP ViT-B/32, DINO ViT-S/16, SBERT), 14 vision datasets (CIFAR-10/100, STL-10, SVHN, MNIST, Fashion-MNIST, BloodMNIST, DermaMNIST, PathMNIST, and 5 ImageNet semantic subsets: animal, vehicle, artifact, food, scene), and 19 SBERT text corpora (AG News, 20 Newsgroups, DBpedia, IMDB, Yelp subsets). Total: 427 pairs (91 ResNet-50, 55 ViT-B/16, 55 CLIP, 55 DINO, 171 SBERT). The 5 ImageNet subsets are constructed from ImageNet-1K validation images grouped by WordNet semantic hierarchy, with $N=2000$ samples each.

Results.

Key findings. (i) At 427 evaluated pairs the overall Pearson correlation is $r=0.970$, compared with $r=0.928$ in E1c. Because pairs share datasets and the larger set is not an independent replication, this increase does not validate a theorem. (ii) All 20 ImageNet-only pairs are superadditive, and all 30 natural×ImageNet cross-domain pairs are superadditive; this is an empirical property of these extracted features. (iii) ResNet-50 achieves 100% superadditivity across 91 pairs (14 datasets including medical and ImageNet) on this benchmark. In contrast, CLIP ViT-B/32 and DINO ViT-S/16 achieve only 31% and 38% superadditivity respectively. Analysis reveals a clear pattern: within-domain pairs (natural×natural, ImageNet×ImageNet) achieve near-perfect superadditivity (CLIP: 90%

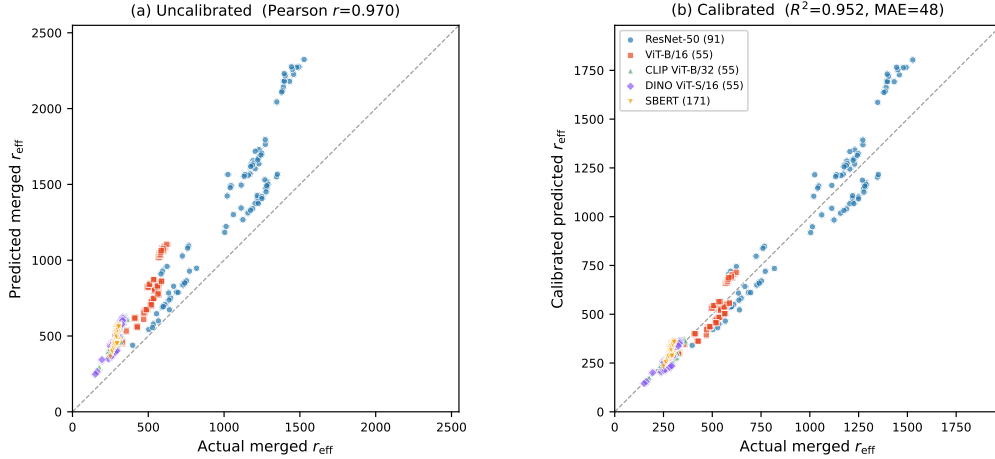


Figure 5: Predicted vs. actual merged r_{eff} for all 427 pairs across 5 encoder families. (a) Uncalibrated: predictions are systematically high but strongly correlated (Pearson $r=0.970$). (b) After per-probe calibration: $R^2=0.952$, $\text{MAE}=48$. Color encodes encoder family; the identity line is dashed.

Table 10: E1d: Historical Collapse-4S association study (427 pairs, 5 probes).

Probe	n	Superadd. %	Pearson r	c	R^2_{cal}
ResNet-50 (14 datasets)	91	100%	0.940	0.776	0.760
ViT-B/16 (11 datasets)	55	69.1%	0.898	0.647	0.487
CLIP ViT-B/32 (11 datasets)	55	29.1%	0.950	0.595	0.839
DINO ViT-S/16 (11 datasets)	55	36.4%	0.913	0.584	0.554
Vision (all probes)	256	64.5%	0.965	per-probe	0.942
+ SBERT text (19 corpora)	+171 (427)	78.2%	0.970	per-probe	0.952

on ImageNet-only, 100% on natural-only; DINO: 100% on both), while cross-domain pairs (handwritten×ImageNet, natural×digit) fail for nearly all evaluated pairs. This suggests that self-supervised encoders learn domain-specific spectral structures with heterogeneous alignment and spectral coupling discarded by the global summaries—an important boundary condition for the Collapse Model’s applicability. (iv) On 171 SBERT text pairs the correlation drops to $r=0.793$ (98.8% superadditive). The narrow r_{eff} range (IQR: 285–298) may contribute, but these results do not establish cross-modal generalization. (v) Calibration constant varies by encoder family ($c \in [0.57, 0.78]$), but a single global $c=0.712$ achieves $R^2=0.898$ across all 158 vision pairs, describing the average bias on this benchmark.

Understanding non-superadditive pairs. ViT-B/16 achieves only 69% superadditivity (38/55), lower than ResNet-50’s 100% (91/91). Analysis of the 17 ViT failures reveals a clear pattern: all failures occur in the energy-imbalance regime ($\gamma \gg 1$ or $\gamma \ll 1$). At $\gamma \geq 3$, 80% of pairs fail (8/10); at $\gamma \in [0.5, 1.5]$, only 4% fail (1/24). The failures concentrate in handwritten×ImageNet cross-domain pairs (10/10 fail, 100%)—these combine low-rank handwritten features ($r_{\text{eff}} \approx 140$ for MNIST via ViT-B/16) with high-rank ImageNet features ($r_{\text{eff}} \approx 600$), creating $\gamma \in [2.0, 4.6]$ far beyond the balanced regime where the two-component spectral mixture is accurate. At large scale (55 pairs each), CLIP ViT-B/32 and DINO ViT-S/16 show only 29% and 36% superadditivity overall, with failures concentrating in cross-domain pairs: handwritten×ImageNet (100% fail), natural×digit (100% fail), natural×handwritten (100% fail). Within-domain pairs (natural-only, ImageNet-only) remain near-perfect.

Root cause: bimodal principal angles. Comparing the same dataset pair across encoders reveals that the failure is not caused by energy imbalance (γ) but by a structural violation of the uniform- α assumption. Three mechanisms interact:

(a) Systematically higher alignment. On the same pair (e.g., CIFAR-10×ImageNet-animal), CLIP’s subspace alignment $\alpha=0.41$ is $2.7\times$ higher than ResNet-50’s $\alpha=0.15$. Contrastive pre-training (CLIP) and self-distillation (DINO) push cross-domain features into a shared semantic subspace, creating high directional overlap where ResNet-50 preserves domain-specific directions.

(b) Extreme spectral concentration. CLIP/DINO feature spectra are highly non-uniform: $\kappa=15,000\text{--}1,200,000$ vs. ResNet-50’s $\kappa=70\text{--}400$. This concentration makes average alignment less informative because overlap on a few high-mass directions can dominate many complementary tail directions.

(c) Bimodal principal-angle distribution. The individual principal angles $\cos^2 \theta_i$ are not approximately uniform (as the model assumes) but bimodal: the top-5 directions are nearly aligned ($\overline{\cos^2 \theta} \approx 0.94$ for CLIP on CIFAR-10×STL-10) while the bottom directions remain orthogonal ($\overline{\cos^2 \theta} \approx 0.01$). The Collapse Model averages these into $\bar{\alpha}=0.52$, predicting moderate superadditivity. In reality, the dominant (high- σ) directions are fully redundant and the orthogonal (low- σ) directions carry negligible energy, so the merge provides almost no diversity gain ($\Delta r_{\text{eff}}=31$ vs. ResNet-50’s $\Delta r_{\text{eff}}=282$ on the same pair).

This analysis identifies a precise applicability boundary: the Collapse Model is reliable when the principal-angle distribution is approximately uniform (supervised encoders, within-domain pairs), but fails when it is bimodal (self-supervised encoders on cross-domain pairs). A natural fix—replacing the uniform average $\bar{\alpha} = \text{SA}_k$ with an energy-weighted alignment $\bar{\alpha}_w = \sum_i (\sigma_i / \|H\|_*) \cos^2 \theta_i$ —would down-weight orthogonal low-energy directions and up-weight the redundant high-energy ones, potentially restoring predictive accuracy for self-supervised encoders. However, our experimental validation shows that this correction does not improve predictive accuracy: on CLIP ($n=61$), the uniform SA_k yields Pearson $r=0.951$, MAE=206, while $\bar{\alpha}_w$ yields $r=0.935$, MAE=218 (worse); DINO shows a similar pattern (r drops from 0.912 to 0.904, MAE increases by 4%); ResNet-50 and ViT-B/16 also show no improvement. A possible explanation is that simple energy weighting over-emphasises the first singular direction’s redundancy ($\cos^2 \theta_1 \approx 0.87$ for that direction), while improperly suppressing the estimated contribution of intermediate directions to the merge. More refined corrections (e.g., piecewise paired-direction models, per-direction threshold truncation, or adaptive k selection) may break through this bottleneck and are left to future work.

M.7 Calibration: From Systematic Bias to Quantitative Prediction

The Collapse Model systematically over-predicts on the evaluated real-data pairs (Remark 4). The log-space over-prediction is approximately constant on the evaluated probes and domains (mean=0.293, std=0.164), so we test whether a multiplicative constant c (Eq. 24) improves within-benchmark calibration. This does not guarantee out-of-encoder quantitative prediction.

Cross-dataset generalization. To verify that c is not dataset-specific, we perform held-out calibration: estimate c on one experiment and test on another (Table 11).

Table 11: Cross-dataset calibration. c is estimated on the calibration set and applied (without re-fitting) to the test set. All setups dramatically improve over the uncalibrated baseline ($R^2 < 0.44$).

Calibration set	Test set	c	n_{test}	R^2	RMSE
E1b (30 pairs)	E1c (57 pairs)	0.768	57	0.859	86.3
E1c (57 pairs)	E1b (30 pairs)	0.746	30	0.888	80.7
E1c (57 pairs)	E_diverse (30 pairs)	0.746	30	0.897	82.8
E1b (30 pairs)	E_diverse (30 pairs)	0.768	30	0.894	83.7
LOO-CV (all 111 pairs, original)		0.736	111	0.877	86.1
LOO-CV (158 vision pairs, E1d)		0.712	158	0.896	121.7
Global (all 427 pairs, E1d)		0.650	427	0.920	105.0

Per-probe calibration (Table 12). The calibration constant varies systematically across encoder families, reflecting differences in spectral structure:

Table 12: Per-probe calibration on the 427-pair dataset (E1d). Each probe is calibrated independently. The variation in c is monotone with the encoder’s effective spectral flatness.

Probe	n	Pearson r	c	R^2_{cal}	Superadd. %
ResNet-50	91	0.940	0.776	0.760	100%
ViT-B/16	55	0.898	0.647	0.487	69%
CLIP ViT-B/32	55	0.950	0.595	0.839	29%
DINO ViT-S/16	55	0.913	0.584	0.554	36%
Vision (global)	256	0.965	per-probe	0.942	65%

Encoders with higher effective rank relative to sample size require stronger correction (c further from 1), consistent with the finite-sample upper-bound mechanism: ResNet-50 features ($d=2048$, $r_{\text{eff}} \approx 350$) are spectrally sparser than ViT features ($d=768$, $r_{\text{eff}} \approx 400$), so the two-component spectral mixture approximation incurs less error. A global $c=0.712$ across all vision probes yields $R^2=0.898$ on the same benchmark; out-of-family validation is required before using a shared constant.

Figure 4 shows the E1c scatter plot before and after calibration.

M.8 E_diverse: Cross-Architecture Probe Independence

Design. To test whether the Collapse Model’s predictions are robust to the choice of encoder, we evaluate the same 6 dataset pairs (from E1b) using both ResNet-50 and ViT-B/16 as independent probes, giving 12 total configurations.

Results. ResNet-50: $R^2 = 0.557$, Pearson $r = 0.9987$, 100% superadditive. ViT-B/16: $R^2 = -16.1$ (systematic over-prediction), Pearson $r = 0.9825$, 83.3% superadditive. Cross-architecture Pearson $r \geq 0.982$ on these six pairs suggests similar directional rankings, but cannot establish probe independence. The Pearson vs. R^2 gap for ViT-B/16 shows encoder-dependent scale bias despite stable ordering.

M.9 E4: Cross-Probe Consistency

Design. We compute the 6×6 pairwise SA_k alignment matrix M independently using ResNet-50 and ViT-B/16 on 4 common datasets {CIFAR-10, MNIST, STL-10, SVHN}. The 6 upper-triangle entries of each matrix are compared via Pearson r .

Results. Cross-probe Pearson $r = 0.891$ ($n = 6$, $p = 0.017$). Both probes agree on the direction of domain distance for all 6 pairs. CIFAR-10/STL-10 are close in both; CIFAR-10/MNIST are far in both. This small- n agreement does not separate dataset geometry from shared encoder biases and is reported only as a consistency check.

M.10 E_downstream: Effective Rank as Downstream Quality Proxy

Motivation. The Collapse Model predicts r_{eff} of merged features, but does higher r_{eff} actually correspond to better downstream performance? We test this by computing r_{eff} and linear-probe (LP) accuracy for 10 pretrained encoders on 5 datasets (50 total pairs).

Design. Encoders: ResNet-{50, 101}, EfficientNet-B4, ConvNeXt-T, DenseNet-121, ViT-B/16, DeiT-B, Swin-T, BEiT-B, RegNet-Y (covering CNN, ViT, and hybrid families). Datasets: CIFAR-10, CIFAR-100, STL-10, SVHN, Fashion-MNIST. For each (encoder, dataset) pair: extract $N=2000$ train features, compute r_{eff} ; train LP (sklearn LogisticRegression) on train, evaluate on test. Report per-dataset Spearman $\rho(r_{\text{eff}}, \text{LP acc})$ across 10 encoders.

Results.

Dataset	All 10 probes		ViT family ($d=768$)		CNN family ($d\geq 1024$)	
	n	ρ	n	ρ	n	ρ
CIFAR-10	10	+0.64	5	+0.90	5	+0.30
CIFAR-100	10	+0.61	5	+0.90	5	+0.30
STL-10	9	+0.73	5	+0.90	4	+0.20
SVHN	10	-0.03	5	+0.10	5	-0.30
Fashion-MNIST	10	+0.03	5	+0.60	5	-0.10

The “all probes” column ($\rho \approx +0.59$ on natural images) is confounded by feature dimension: CNNs ($d=2048$) have systematically higher r_{eff} than ViTs ($d=768$), inflating the apparent correlation. Controlling for dimension reveals a sharper picture: within the ViT family ($d=768$, 5 probes), r_{eff} is strongly predictive of LP accuracy on natural images ($\rho=0.90$ on CIFAR-10, CIFAR-100, and STL-10), while within the CNN family ($d\geq 1024$, 5 probes) the correlation is weak ($\rho=0.20$ – 0.30). This suggests that r_{eff} captures useful discriminative information primarily for architectures with compact feature spaces ($d<1024$), where spectral diversity is more tightly coupled to representation quality.

Implication for the Collapse Model. r_{eff} should be understood as a measure of representational diversity—the number of independent directions activated by the data—rather than a direct proxy for downstream accuracy. Collapse-4S predicts when merging may increase this spectral diversity. Within one evaluated architecture family, diversity and accuracy are positively correlated ($\rho=0.90$ for ViTs), but this association neither establishes necessity nor guarantees that r_{eff} -based selection improves a new task. Across architecture families or across datasets of varying difficulty, additional task-specific metrics are needed.

Sample-size sensitivity. Empirically, the over-prediction ratio grows with N/r_{eff} on 10 representative ResNet-50 pairs: at $N=200$ the mean ratio is 1.021, rising monotonically to 1.545 at $N=2000$ (Spearman $\rho=0.80$ between N/r_{eff} and log bias). This association helps predict calibration variation across encoders (ResNet-50 $c=0.78$, DINO $c=0.58$), but is not asserted as a theoretical error law.

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1086 The corrected Stage-2 control uses target-cluster bootstrap; oracle minus frozen
1087 identity is -0.0670 with 95% CI $[-0.0977, -0.0440]$. The rank- L coverage statement
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1121 feature/adaptation requirements, device assignment, run counts, and observed
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1157 also preserve, hide, or amplify biases in the encoder and candidate datasets and
1158 may underselect small groups. We therefore require task-aware validation and do
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